HDSoC DAQ Manual

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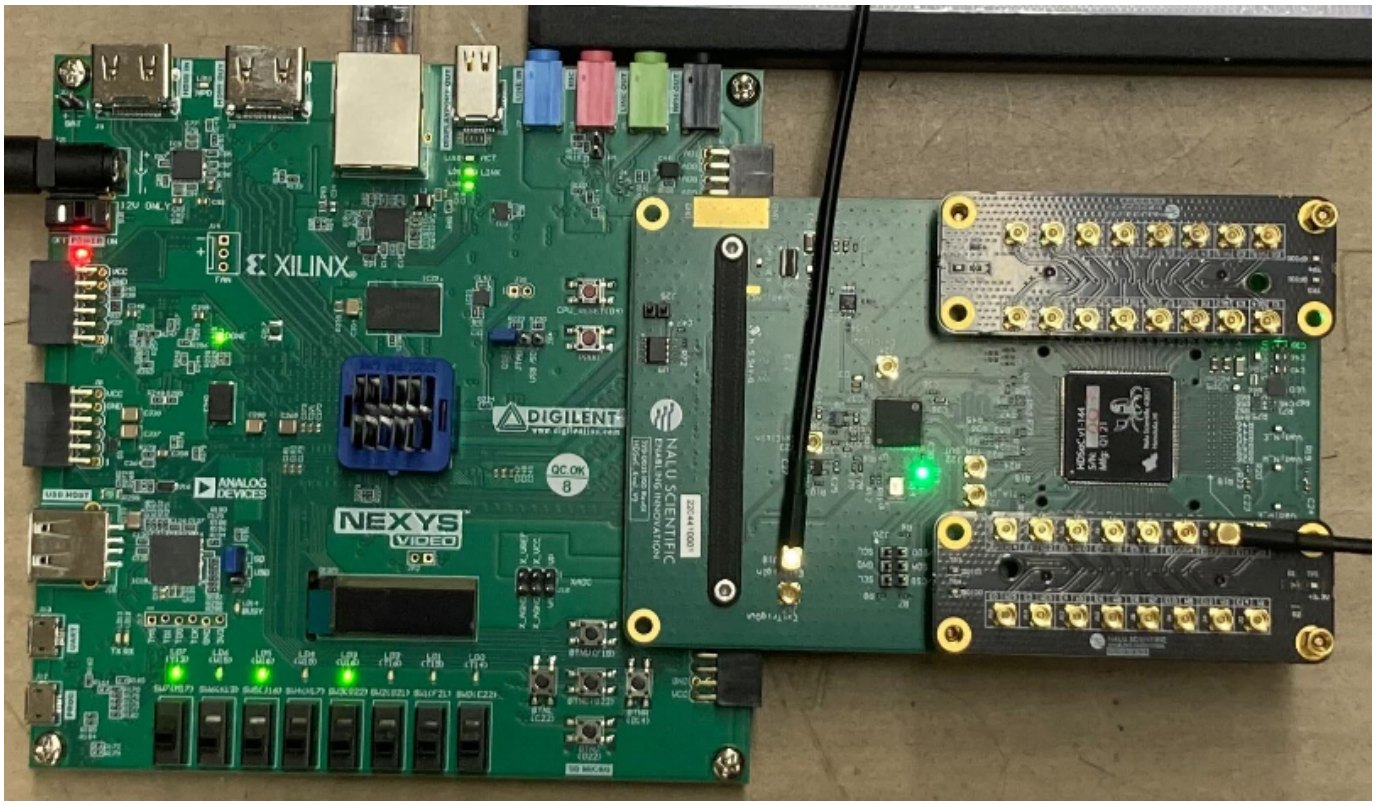
1. Welcome to the HDSoC DAQ Manual



The purpose of this manual is to aid users with setup, usage, and debugging of the HDSoC data acquisition (DAQ) system. This DAQ's purpose is to be used for readout of the Nalu's HDSoC FMC boards (and possibly other boards). Most topics are simplified to only include information needed for operating this DAQ. Some external links are provided for additional, generalized information.

Many of the guides on this webpage are thorough, as they are aimed to give solutions to problems I've encountered. However, every system is different; there may be some additional debugging to be done on the user's end.

1.1 PDF Version

A pdf version of this manual is automatically generated using MkDocs with pdf plugin.

1.2 Contact

Manual Written by Jack Carlton.
Ph.D. Candidate, Department of Physics and Astronomy, University of Kentucky.
Email: j.carlton@uky.edu
GitHub: jaca230

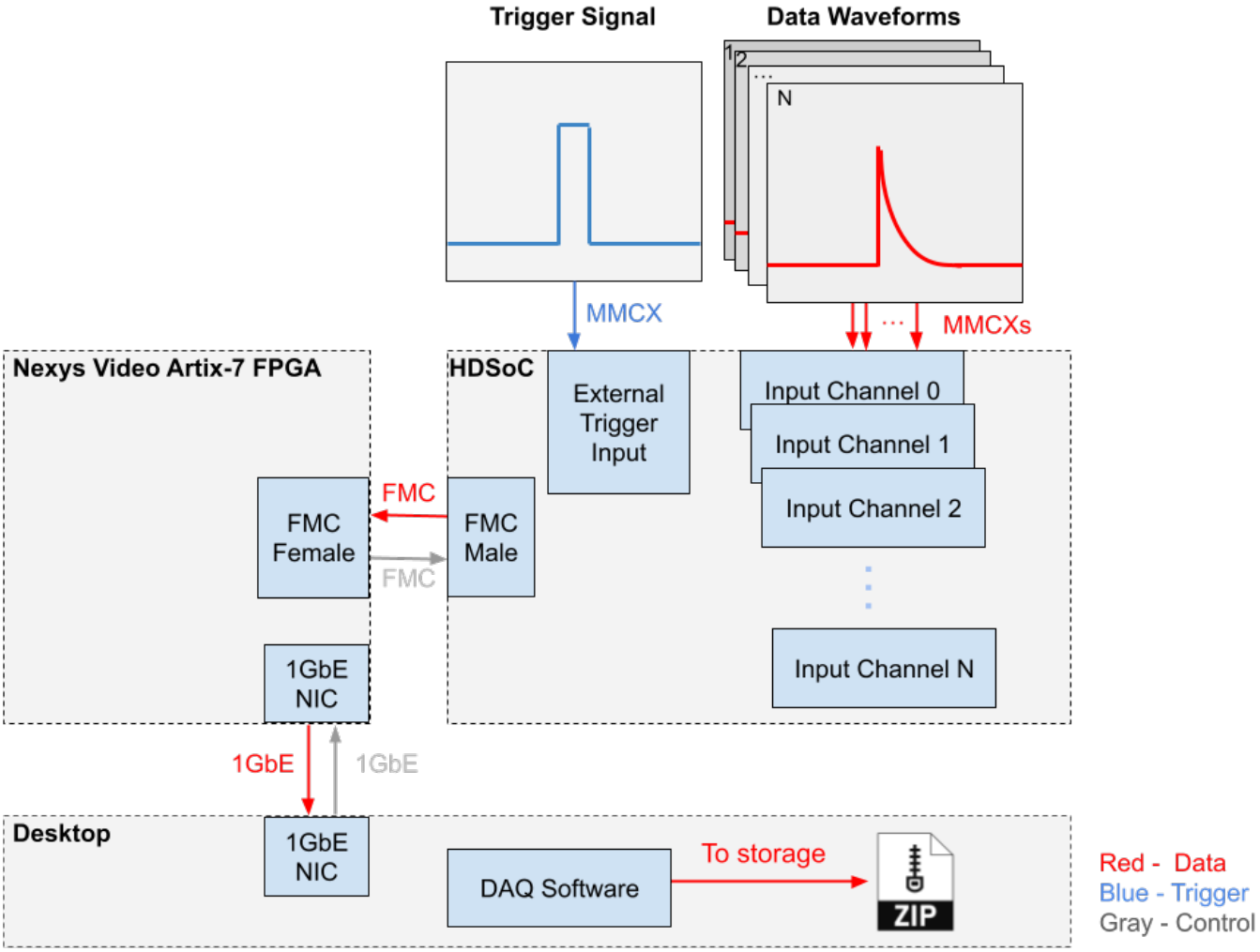
Feel free to reach out with any questions, to correct mistakes, point out missing information, or otherwise. Feel free to contribute additional information on github or otherwise.

Last update: September 9, 2025

2. Hardware

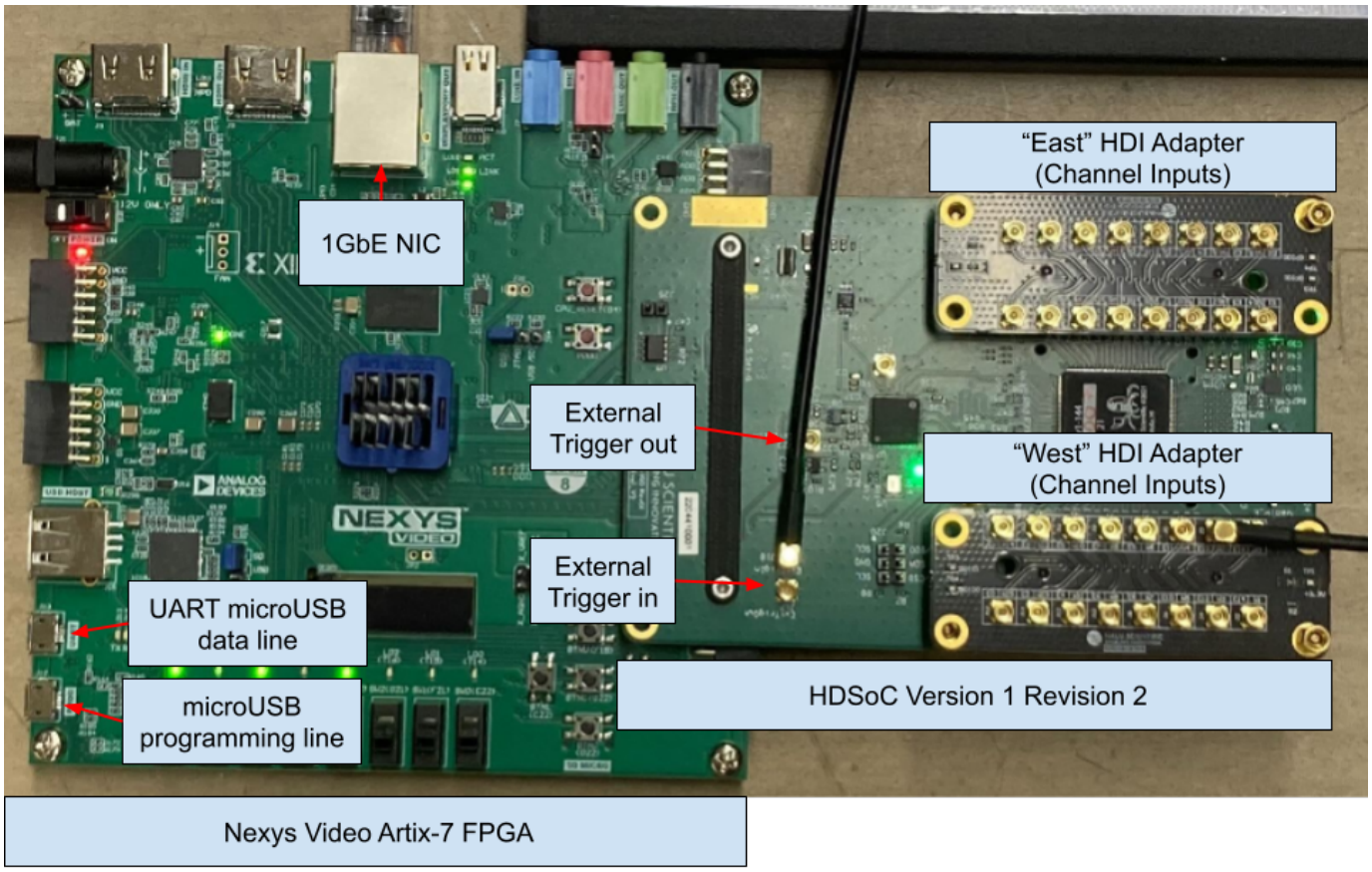
2.1 General Hardware Overview

2.1.1 Conceptual Diagram



- **Trigger Signal**: An external signal to tell the board when to digitize the input waveforms. Should be LVCMOS25 (0-2.5V) with width 10-100 ns.
- **Waveforms**: Analog signal to be digitized. Should be LVCMOS25 (0-2.5V). Max width is 1984 nanoseconds by default, but this depends on the sampling rate (default 1Gsps).
- **HDSoC**: Digitization board that handles inputs across multiple channels. Can be set to trigger on external or internal (self) signals.
- **Nexys Video Artix-7 FPGA**: Acts as a parent board to the HDSoC. Handles 1GbE communication and firmware storage among other things.
- **Desktop**: Receives and processes events that are effectively streamed from the digitization system.

2.1.2 Labled Picture



2.2 1GbE NIC (Gigabit Ethernet Network Interface Card)

2.2.1 Overview

These NICs are generally PCIe Cards that are "plug and play". They provide a 1 gigabit per second ethernet connection for the host computer.

2.2.2 Configuration

If you machine has a GUI, you may find it easier to edit network settings that way. Otherwise, you can edit settings from command line. For example for UKY's teststand we use:

```
nmcli connection modify enp3s0 \
    ipv4.addresses 192.168.1.1/24 \
    ipv4.method manual \
    connection.autoconnect yes \
    ipv6.method ignore
```

You may need to create a connection configuration file first if it doesn't exist. For example for UKY's teststand we use:

```
nmcli connection add type ethernet con-name enp3s0 ifname enp3s0
```

In particular, the `ipv4.addresses` is important. Here the port is specified to accept any traffic on the 192.168.1.xxx subnet. See the networking page for more details.

2.3 Nexys A7 Video Card

2.3.1 Overview

The Nexys A7 is a versatile and high-performance FPGA development board from Digilent, featuring the Xilinx Artix-7 FPGA. It is like the "mother board" for the HDSoC. For this use case it will provide power, an interface to update firmware, and an interface for data transfer. You can read more about this board on diligent's Nexys Video Artix-7 FPGA page.

2.3.2 FMCs

HDSoC

The HDSoC (or other nalu board) connects to the FPGA board via an FMC connector. See HDSoCv1 rev 2 for more info.

2.3.3 Wired Connections

USB A to microUSB

A USB A to microUSB can be used for UART communication. This can be used for data transfer via UART (very slow, limited to ~100 KB/s) and uploading firmware to the board. See the hardware diagram for port locations.

1GbE Connection

Connect an ethernet cable between the board and a host desktop computer. Most modern ethernet cables should work. See the hardware diagram for port locations.

2.3.4 Configuration

To congfigure the board, the appropriate firmware must be uploaded to it. This can be retrieved from nalu scientific's board downloads page. There are several options, the easiest two are:

1. Use microSD card with loaded firmware (see HDSoC quickstart guide)
2. Upload the firmware via Vivado

Steps to complete step 2 are below.

Downloading the firmware

Navigate to Nalu Scientific's support website. Go to the page for the HDSoCv1-evalr2 (or appropriate board). Download the most recent firmware version. This DAQ has been tested with firmware version 938. The file should be something like `HDSoC_eval_v938.bit`.

Downloading Vivado

Note: Vivado may take a *long* time to install because it is a very large program. This step could take serveral hours. A full Vivado installation can be around 200 GB. Below is a "minimal" installation that is ~50GB

The easiest way to upload firmware to the board is via Vivado. Download the most recent version of Vivado for your system from Xilinx's download page. You will need to make an account with AMD to install. This is easiest on a machine with a GUI. For a simple install follow these steps:

1. Run installer. On linux run the bin file with `/path/to/.../installer.bin`.
2. On the welcome page, hit next
3. Enter credentials, download and install now, hit next
4. Under products to install select Vivado, hit next
5. For customizing the installation.

a. Under "Design Tools" select only "Vivado Design Suite"

b. Under "Devices" select only "Production Devices">"7 Series"

c. Under "Installation Options", make sure you have "Install Cable Drivers" selected. Hit next.

6. Agree to License agreements, hit next.
7. Select which directory to install to. Hit next. Begin installation.

Uploading Firmware

Once Vivado is installed run it. On linux you may need to source the settings file with `source /path/to/.../Xilinx/Vivado/202x.x/settings64.sh`, then you can run the command `vivado` to launch vivado.

A new project needs to be created. This can be a "dummy" project since we're just uploading firmware. Step 1 of this guide gives all the details needed to createae new project.

To load firmware onto the Nexys A7 using Vivado, plug the USB A to microUSB connection into the program port of the Nexys A7 board. See the hardware diagram for port locations. Follow these steps to upload firmware such that it persists when the board reboots:

1. **Open Vivado Hardware Manager and Connect to the Board**
2. Open Vivado and go to **Tools > Hardware Manager**. **Hardware Manager** should also be visible on the left sidebar of the project page.
3. In the **Hardware Manager** window, click **Open Target** and select **Auto Connect** or choose the appropriate JTAG or USB connection to connect to your board. If the board is connected properly, Vivado should automatically detect it.
4. **Convert the Bitstream to a Binary File** To program the flash, you need to convert the bitstream to a binary file. Nalu provides `.bit` files, but we can convert the `.bit` file into a `.bin` file. Here's how you can do it:
5. After the bitstream is generated, open the TCL Console (**Window > Tcl Console**) and use the following command to generate the `.bin` file from the `.bit` file:

   ```
   write_cfgmem -force -format bin -interface SPIx4 -size 32 -loadbit "up 0x0 /path/to/input/firmware.bit" -file /path/to/output/firmware.bin
   ```

   Replace the paths with the appropriate file locations for your system.

   Example:

   ```
   write_cfgmem -force -format bin -interface SPIx4 -size 32 -loadbit "up 0x0 /home/pioneer/vivado_stuff/Nexsys_Video_A7_For_HDSoc/HDSoC_eval_v938.bit" -file /home/pioneer/vivado_stuff/Nexsys_Video_A7_For_HDSoc/HDSoC_eval_v938.bin
   ```
6. **Add Configuration Memory Device** Once the `.bin` file is ready, go back to the **Hardware Manager** in Vivado and:
7. Right-click on your connected device in the **Devices** pane (in the Hardware Manager).
8. Click **Add Configuration Memory Device**.
9. In the dialog box, select the appropriate part for your flash device. For the Nexys A7, the part number will likely be something like **S25FL256Sxxxxxx0-SPI-x1_x2_x4**.
10. Click **OK** to confirm.
11. **Program the Flash Memory** After selecting the configuration memory device:
12. Vivado will prompt you to **program the flash memory**.
13. Select the **firmware `.bin` file** you generated earlier.
14. Click **Program** to start the flashing process. The process will take some time depending on the size of your firmware file and the connection speed.
15. **Power Cycle the Device** After programming is complete, power down the device and power it back on. Once powered on, you should see indicators (such as green lights on the HDSoC board) signaling that the firmware has been loaded successfully.

2.4 HDSoCv1 rev 2

2.4.1 Overview

The HDSoC is a high density digitizer system on a chip evaluation board. It provides a flexible digitization system. The high density part makes it appealing to for use in the ATAR DAQ, as the ATAR will have thousands of channels. You can read more in the HDSoC product sheet or on Nalu Scientific's HDSoC page.

2.4.2 Wired Connections

MMCX connector

The HDSoC's inputs (channel inputs and external trigger) and outputs (external trigger output) use a MMCX connectors. See the labeled picture and conceptual diagram to see where these inputs are on the board.

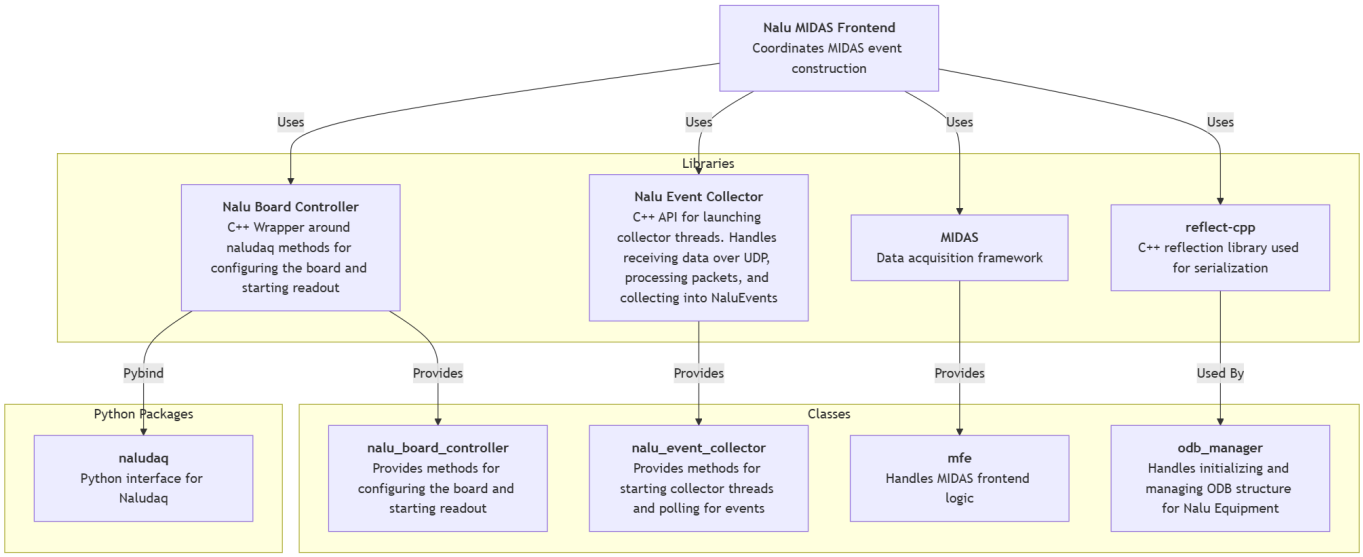
2.4.3 Configuration

See the HDSoC quickstart guide. You don't need to worry about using the SD card to boot the firmware if you've already uploaded it.

Last update: May 13, 2026

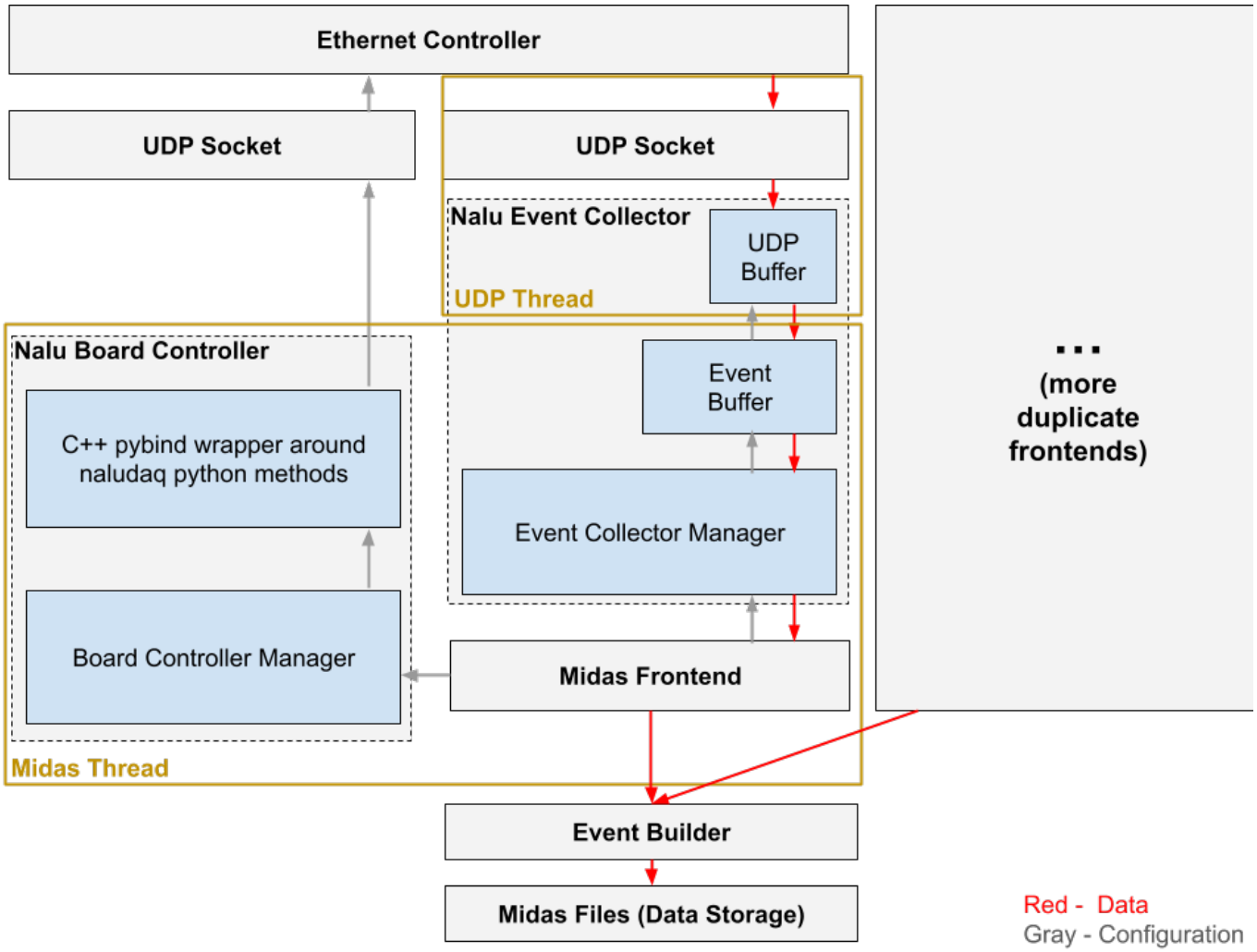
3. General Sofware Overview

3.1 Software Dependency Diagram



Note: an SVG version of this diagram with links to each repository is avaiable here.

3.2 Software Data and Control Flow Diagram



- **Ethernet Controller**: The systems ethernet controller which sockets are formed on.
- **UDP Socket**: Two sockets are constructed. One by the Python board-control stack to communicate with the board and one by the UDP receiver to receive data.
- **Nalu Event Collector**: A library to collect events from a UDP socket receiving events from a nalu scientific board.
- **UDP Buffer**: Receives UDP packets and puts them in a buffer to be processed.
- **Event Buffer**: Processes UDP packets into events and puts them in a buffer.
- **Event Collector Manager**: Manages interfacing with the buffers. I.e. getting data, configuration, etc.
- **Nalu Board Controller**: C++ methods to configure the nalu scientific board to prepare for data taking.
- **C++ pybind wrapper**: C++ wrapper around the Python board-control stack.
- **Board Controller Manager**: Manages interfacing with the board from another C++ program.
- **Midas Frontend**: Handles run control, configuration via ODB, data bank creation, etc. Interfaces with the board controller and event collector.
- **Event Builder**: Builds events from data banks provided by potentially multiple frontends. Not necessary if only using one frontend.

3.3 HDSOC_DAQ

3.3.1 Overview

The HDSoC DAQ software is a MIDAS frontend that interfaces with several other softwares (see software dependencies page) to read out events from a nalu scientific board while working within the MIDAS framework.

The installation chain is now split into two parts:

- External dependencies that must already exist on the host, especially MIDAS and optionally ROOT.
- Dependencies that are handled by this repository itself, either through a repo-local Python virtual environment or through CPM during the CMake configure step.

3.3.2 Installation

Make sure you have installed the development tools, the Python packages, and MIDAS before continuing. If you need private repository access, also make sure you have set up access to the PIONEER experiment repository.

1 **Clone the repository:**

```
git clone git@github.com:PIONEER-Experiment/hdsoc_daq.git
cd hdsoc_daq
```

2 **Set up the environment**

The main shell entrypoint is now:

```
source ./scripts/environment/activate_environment.sh
```

This helper will:

- detect and save `MIDASSYS`, `MIDAS_EXPTAB`, and `MIDAS_EXPT_NAME` if needed
- create or refresh the repo-local Python environment if needed
- activate the repo-local Python environment
- isolate the runtime from user-site Python packages

If you only want to detect and save the MIDAS-related variables without activating the Python environment, run:

```
./scripts/environment/helpers/detect_midas_environment.sh
```

This writes:

```
scripts/environment/helpers/environment_variables.sh
```

Check that file to make sure it contains the correct paths. For example:

```
export MIDASSYS=/home/pioneer/packages/midas
export MIDAS_EXPTAB=/home/pioneer/packages/online/exptab
export MIDAS_EXPT_NAME=ATAR_DAQ
export ATAR_DAQ_DIR=/home/pioneer/packages/experiments/hdsoc_daq
```

If you want to create or refresh only the repo-local virtual environment, run:

```
./scripts/environment/helpers/setup_venv.sh
```

3 **Build**

Once `MIDASSYS` points to a working MIDAS install, build with:

```
./scripts/build.sh
```

If you want to force a clean reconfigure first:

```
./scripts/build.sh --overwrite
```

During the CMake configure step, CPM automatically fetches and configures several C++ dependencies, including:

- `nalu_event_collector`
- `nalu_board_controller`
- `reflect-cpp`
- `spdlog`
- `nlohmann/json`
- `pybind11`

So these no longer need to be installed manually as part of the normal `hdsoc_daq` setup flow.

3.3.3 Running

Starting a Midas Webpage

Midas provides a great user interface via their webpage. To start it, use the helper script:

```
./scripts/webpage_scripts/start_midas_webpage.sh
```

Then navigate to `localhost:8080` in your favorite web browser.

Note: If the webpage doesn't appear, manually run `mhttpd` to debug the error output.

Manually Starting the Frontend

I recommend doing this the first time to make sure everything is working properly.

```
./scripts/run.sh -i 0
```

Some useful options are:

```
./scripts/run.sh --build -i 0
./scripts/run.sh --debug -i 0
./scripts/run.sh --setup-venv -i 0
./scripts/run.sh --system-python -i 0
```

Starting the Frontend as a Screen

Use the screening helper script:

```
./scripts/screen_control/screen_frontend.sh -i {index}
```

Note: Excluding the `-i` flag sets the index to `0`. This is the frontend index used to support running multiple frontends.

To stop the screen:

```
./scripts/screen_control/stop_screen.sh -i {index}
```

Starting the Frontend as Midas Program

See the g-2 modified DAQ Manual's guide for adding program startup scripts. For the start command, use the screen command above.

3.3.4 Configuration

See the ODB configuration page for a description of each ODB setting.

Last update: May 22, 2026

4. Software Dependencies

4.1 PIONEER Repository

Some installations are on the PIONEER repository, which requires access to pull from. See how to get access.

4.2 Development Tools

4.2.1 Overview

These tools include compilers, CMake, Python development support, and other utilities that facilitate software development and installation.

4.2.2 Installation Guide

This guide should work for ALMA9. You can use `dnf` for ALMA9, but I prefer to work with `yum`.

1 **Install yum package manager**

```
sudo dnf install yum
```

2 **Update the package index**

```
sudo yum update
```

3 **Enable the EPEL repository**

```
sudo yum install epel-release
```

4 **Install Development Tools and Dependencies**

```
sudo yum groupinstall "Development Tools"
sudo yum install cmake gcc-c++ gcc screen subversion binutils libX11-devel libXpm-devel libXft-devel libXext-devel python3 python3-devel python3-pip
```

5 **Check the CMake version**

```
cmake --version
```

`hdsoc_daq` currently requires CMake `>= 3.23`.

If the installed CMake version is too old, follow these steps to install a newer version manually:

```
wget https://github.com/Kitware/CMake/releases/download/v3.29.0/cmake-3.29.0.tar.gz
tar -xvzf cmake-3.29.0.tar.gz
cd cmake-3.29.0
./bootstrap
make -j$(nproc)
sudo make install
```

Always check Kitware's website for the current version before copying the version number above.

4.3 Python Packages

4.3.1 Overview

The DAQ still depends on a Python board-control stack even though the frontend itself is written mostly in C++. The current setup uses a repo-local virtual environment to hold the Python runtime needed by the board controller path.

4.3.2 Installation Guide

The recommended way to set this up is with the repo helper:

```
./scripts/environment/helpers/setup_venv.sh
```

This creates or refreshes a virtual environment at:

```
.atar_daq_venv
```

and installs the requirements from:

```
requirements/board_tools.txt
```

At the moment, that requirements file includes:

```
naludaq==0.36.0
naluconfigs==18.0.1
```

If you are not using the repo-local virtual environment, you will need those packages available in the Python environment used at runtime.

4.4 ROOT

4.4.1 Overview

ROOT is an open-source data analysis framework developed by CERN. It is needed for some MIDAS features.

4.4.2 Installation Guide

General installation guides are provided by ROOT at their Installing ROOT and Building ROOT from source pages.

Using `yum` Package Manager

1 **Enable the EPEL repository**

```
sudo yum install epel-release
```

2 **Download and Install ROOT**

```
sudo yum install root
```

Building from source

1 **Example building latest stable branch from source**

```
git clone --branch latest-stable --depth=1 https://github.com/root-project/root.git root_src
mkdir root_build root_install && cd root_build
cmake -DCMAKE_INSTALL_PREFIX=../root_install ../root_src
cmake --build . --target install -j4
source ../root_install/bin/thisroot.sh
```

Note: Adjust the ROOT version and the download URL as needed. Always check for the latest version on the official ROOT website.

4.5 Midas

4.5.1 Overview

Midas is a data acquisition system used in high-energy physics experiments. Midas provides the following functionalities:

- Run control
- Experiment configuration
- Data readout
- Event building
- Data storage
- Slow control
- Alarm systems
- ... much more ...

4.5.2 Installation Guide

For a general MIDAS installation, you can follow this Linux Quick Start Guide.

`hdsoc_daq` expects `MIDASSYS` to point to a working MIDAS install before configuration. In particular, the build checks for:

- `$MIDASSYS/include`
- `$MIDASSYS/lib/libmidas.a`
- `$MIDASSYS/lib/libmfe.a`

For the g-2 modified DAQ, we use a custom version of MIDAS, which can be cloned and installed as follows:

1 **Set experiment name environment variable**

```
export MIDAS_EXPT_NAME=ATAR_DAQ
```

2 **Create exptab file**

```
mkdir online
cd online
touch exptab
echo "$MIDAS_EXPT_NAME $(pwd) system" >> exptab
export MIDAS_EXPTAB=$(pwd)/exptab
```

3 **Install MIDAS**

```
cd ..
mkdir packages
git clone --recursive git@github.com:PIONEER-Experiment/midas-modified.git midas
cd midas
mkdir build
cd build
cmake ..
make -j$(nproc) install
cd ..
```

4 **Set `MIDASSYS` environment variable and add to path**

```
export MIDASSYS=$(pwd)
export PATH=$PATH:$MIDASSYS/bin
```

Note: You can hardcode the environment variables `MIDASSYS`, `MIDAS_EXPTAB`, and `MIDAS_EXPT_NAME` by adding the appropriate commands to your `.bashrc` file. The repo also provides environment helpers documented on the scripts page.

5 **(Optional) it is recommended to also install the MIDAS Python package**

```
pip install -e $MIDASSYS/python --user
```

4.6 CPM Managed Libraries

4.6.1 Overview

Several dependencies used by `hdsoc_daq` are no longer meant to be installed manually as part of the normal setup flow. They are fetched automatically by CPM during the CMake configure step.

These include:

- `nalu_event_collector`
- `nalu_board_controller`
- `reflect-cpp`
- `spdlog`
- `nlohmann/json`
- `pybind11`

4.6.2 Installation Guide

No separate manual installation step is needed for the normal `hdsoc_daq` build. Once `MIDASSYS` is set and the Python environment is available, running:

```
./scripts/build.sh
```

will configure CMake and let CPM pull the required C++ packages automatically.

If you want to inspect or build these projects separately, see their upstream repositories:

- nalu_board_controller
- nalu_event_collector
- reflect-cpp

Last update: May 22, 2026

5. Software Add-ons

Last update: March 25, 2025

6. Midas' Online Data Base Configuration Parameters

6.1 ODB Basics

See the g-2 modified DAQ's ODB Basics section for some general MIDAS ODB settings.

6.2 HDSoC DAQ specific ODB Configuration

Note: The HDSoC frontend initializes settings under:

`/Equipment/HDSoC-{frontend #}/Settings`

Some runtime values are derived from other ODB settings rather than stored directly as their own ODB entries. In particular:

- `nalu_event_collector/nalu_udp_receiver/address` and `port` are derived from `nalu_board_controller/nalu_capture/target_endpoint`
- `nalu_event_collector/nalu_event_builder/channels` is derived from enabled channel entries under `nalu_board_controller/nalu_capture/channels`
- `nalu_event_collector/nalu_event_builder/windows` is derived from `nalu_board_controller/nalu_capture/readout_window/windows`
- `nalu_event_collector/nalu_event_builder/trigger_type` is derived from `nalu_board_controller/nalu_capture/trigger/mode`
- `nalu_event_collector/nalu_event_builder/wlc_mode` is derived from `nalu_board_controller/nalu_capture/window_level_control/enabled`

The settings below are the actual ODB entries initialized by the frontend.

6.2.1 Nalu Event Collector

Absolute Window Mask

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/abs_wind_mask`
Description	Bit mask on the absolute window position bits.
Valid Values	Any bit mask, for example `63` = `0x3F`.
Suggested Value	`63`

Channel Mask

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/chan_mask`
Description	Bit mask on the channel index bits.
Valid Values	Any bit mask, for example `63` = `0x3F`.
Suggested Value	`63`

Channel Shift

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/chan_shift`
Description	Global shift in channel indices.
Valid Values	Any non-negative integer.
Suggested Value	`0`

Check Packet Integrity

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/check_packet_integrity`
Description	Whether each received UDP packet should be checked against the expected packet format.
Valid Values	`yes` or `no`
Suggested Value	`no`

Note: Setting this to `no` is the current default in code.

Constructed Packet Footer

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/constructed_packet_footer`
Description	Footer bytes for constructed packets. Potentially useful to consumer programs looking for a specific byte sequence.
Valid Values	Any non-negative 2 byte integer.
Suggested Value	`65535`

Constructed Packet Header

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/constructed_packet_header`
Description	Header bytes for constructed packets. Potentially useful to consumer programs looking for a specific byte sequence.
Valid Values	Any non-negative 2 byte integer.
Suggested Value	`43690`

Event Window Mask

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/evt_wind_mask`
Description	Bit mask on the event window position bits.
Valid Values	Any bit mask, for example `63` = `0x3F`.
Suggested Value	`63`

Event Window Shift

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/evt_wind_shift`
Description	Global shift on event window values.
Valid Values	Any non-negative integer.
Suggested Value	`6`

Packet Size

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/packet_size`
Description	The size in bytes of one raw packet of channel information.
Valid Values	Any positive integer.
Suggested Value	`74`

Start Marker

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/start_marker`
Description	Start marker for a packet of channel information.
Valid Values	Any hexadecimal string.
Suggested Value	`0E`

Stop Marker

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/stop_marker`
Description	Stop marker for a packet of channel information.
Valid Values	Any hexadecimal string.
Suggested Value	`FA5A`

Timing Mask

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/timing_mask`
Description	Bit mask on timing information.
Valid Values	Any non-negative integer representable in the field.
Suggested Value	`4095`

Timing Shift

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_packet_parser/timing_shift`
Description	Global shift in timing information.
Valid Values	Any non-negative integer.
Suggested Value	`12`

UDP Buffer Size

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_udp_receiver/buffer_size`
Description	Number of bytes in the UDP buffer before an overflow condition occurs.
Valid Values	Any positive integer.
Suggested Value	`104857600`

UDP Max Packet Size

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_udp_receiver/max_packet_size`
Description	Maximum number of bytes in a UDP packet that the receiver will consider.
Valid Values	Any positive integer.
Suggested Value	`1040`

UDP Timeout

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_udp_receiver/timeout_sec`
Description	Number of seconds the UDP thread will wait before quitting due to timeout.
Valid Values	Any positive integer.
Suggested Value	`10`

Event Header

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_event_builder/event_header`
Description	Header for each constructed event. Potentially useful to consumer programs looking for specific byte sequences.
Valid Values	Any non-negative 2 byte integer.
Suggested Value	`48059`

Event Trailer

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_event_builder/event_trailer`
Description	Trailer for each constructed event. Potentially useful to consumer programs looking for specific byte sequences.
Valid Values	Any non-negative 2 byte integer.
Suggested Value	`61166`

Max Events in the Event Buffer

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_event_builder/max_events_in_buffer`
Description	Maximum number of events retained in the event buffer. This should be chosen within the RAM constraints of your system.
Valid Values	Any positive integer.
Suggested Value	`10000`

Max Lookback

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_event_builder/max_lookback`
Description	Maximum number of prior events the event builder will look back through when trying to match a packet.
Valid Values	Any positive integer.
Suggested Value	`2`

Note: Look backs only occur shortly after new events begin, in case packets are received out of order.

Max Trigger Time

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_event_builder/max_trigger_time`
Description	The maximum number of clock ticks the board will report before restarting at `0`.
Valid Values	Any positive integer.
Suggested Value	`16777216`

Clock Frequency

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_event_builder/clock_frequency`
Description	Clock frequency of the board in Hz, used to convert clock ticks into time-related quantities during event building.
Valid Values	Any positive integer.
Suggested Value	`23843000`

Time Threshold

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_event_builder/time_threshold`
Description	Maximum number of clock ticks two packets can be separated by and still be considered part of the same event.
Valid Values	Any positive integer.
Suggested Value	`5000`

Event Completion Time

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_event_collector/nalu_event_builder/event_completion_time_us`
Description	The amount of time in microseconds an event will allow new packets to be added before being marked complete. This is mainly relevant for self trigger mode.
Valid Values	Any positive integer.
Suggested Value	`10000`

Note: This also acts as a delay before the first event can complete.

6.2.2 Nalu Board Controller

Channel DAC Value

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/channels/{channel_index}/dac_value`
Description	The digital-to-analog converter value for the channel.
Valid Values	Any non-negative integer.
Suggested Value	`1804`

Note: These values are only applied if Assign DAC Values is set to `yes`.

Channel Enabled

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/channels/{channel_index}/enabled`
Description	Whether that channel is enabled for readout.
Valid Values	`yes` or `no`
Suggested Value	`yes`

Channel Trigger Value

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/channels/{channel_index}/trigger_value`
Description	Threshold used for self triggering on that channel.
Valid Values	Any non-negative integer.
Suggested Value	`0`

Assign DAC Values

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/assign_dac_values`
Description	Whether DAC values are assigned and used by the board.
Valid Values	`yes` or `no`
Suggested Value	`no`

Target Endpoint

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/target_endpoint`
Description	IP address and port that the board will send data to over 1 GbE.
Valid Values	Any valid `host:port` string.
Suggested Value	`192.168.1.1:12345`

Digitization Windows

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/readout_window/windows`
Description	Number of windows, each 32 samples, that will be digitized.
Valid Values	Any positive integer between `1` and `62` for the HDSoC.
Suggested Value	`1`

Note: See page 18 of the NaluScope manual for a better explanation of the windows parameter.

Digitization Lookback

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/readout_window/lookback`
Description	Number of windows, each 32 samples, to look back before the trigger.
Valid Values	Any positive integer between `1` and `62` for the HDSoC.
Suggested Value	`1`

Note: See page 18 of the NaluScope manual for a better explanation of the lookback parameter.

Digitization Write After Trigger

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/readout_window/write_after_trigger`
Description	Number of windows, each 32 samples, to continue digitizing after the trigger event.
Valid Values	Any positive integer between `1` and `62` for the HDSoC.
Suggested Value	`1`

Note: See page 18 of the NaluScope manual for a better explanation of the write-after-trigger parameter.

Digitization Lookback Mode

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/readout_window/lookback_mode`
Description	Lookback mode string used by the board-control layer.
Valid Values	Any string accepted by the board-control software.
Suggested Value	`default`

Trigger Mode

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/trigger/mode`
Description	Trigger mode to use. Common values are external, self, or immediate triggering.
Valid Values	`ext`, `self`, or `imm`
Suggested Value	`ext`

Trigger Low Reference

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/trigger/low_reference`
Description	The lower 4-bit discriminator subrange boundary used by the ASIC trigger circuit. Together with high_reference, this sets the voltage subrange in which the per-channel trigger threshold is evaluated.
Valid Values	Any integer between 0 and 15 inclusive such that the value is less than the high reference
Suggested Value	`0`

Note: See page 21 of the NaluScope manual for a better explanation of the Low and High References.

Trigger High Reference

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/trigger/high_reference`
Description	The upper 4-bit discriminator subrange boundary used by the ASIC trigger circuit. Together with low_reference, this narrows the voltage range for the trigger discriminator, allowing finer thresholding within that subrange.
Valid Values	Any integer between 0 and 15 inclusive such that the value is greater than the low reference
Suggested Value	`15`

Note: See page 21 of the NaluScope manual for a better explanation of the Low and High References.

Trigger Rising Edge

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/trigger/rising_edge`
Description	Whether to trigger or rising or falling edge
Valid Values	`yes` or `no`
Suggested Value	`yes`

Window Level Control Configure

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/window_level_control/configure`
Description	Toggle for whether or not window level control will be configured.
Valid Values	`yes` or `no`
Suggested Value	`no`

Note: If WLC is disabled (below) you often do not want to bother configuring it to "off" every time because this takes extra time to re-intialize the board.

Window Level Control Enabled

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/window_level_control/enabled`
Description	Whether or not window level control mode is enabled.
Valid Values	`yes` or `no`
Suggested Value	`no`

Note: Window level control must be applied on board initialization, you cannot change it in between runs without restarting the frontend.

Window Level Control Reinitialize After Change

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_capture/window_level_control/reinitialize_after_change`
Description	Whether window level control should be reinitialized after a configuration change.
Valid Values	`yes` or `no`
Suggested Value	`yes`

Board Endpoint

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_board/board_endpoint`
Description	IP address and port of the board for control communication.
Valid Values	Any valid `host:port` string.
Suggested Value	`192.168.1.59:4660`

Board Clock File

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_board/clock_file`
Description	Path to a clock file for the board.
Valid Values	Any valid path or empty string.
Suggested Value	`""`

Board Configuration File

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_board/config_file`
Description	Path to a configuration file for the board.
Valid Values	Any valid path or empty string.
Suggested Value	`""`

Host Endpoint

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_board/host_endpoint`
Description	IP address and port of the host used for control communication with the board.
Valid Values	Any valid `host:port` string.
Suggested Value	`192.168.1.1:4660`

ASIC Serial Mode

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_board/asic_serial_mode`
Description	Whether or not the board should use ASIC serial mode during configuration and operation.
Valid Values	`yes` or `no`
Suggested Value	`no`

Board Model

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/nalu_board_controller/nalu_board/model`
Description	Name of the board model used.
Valid Values	Any valid board model string supported by the board-control software.
Suggested Value	`HDSoCv1_evalr2`

6.2.3 Logging Parameters

Event Collector Logging Level

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/logging/event_collector_log_level`
Description	Log level for messages produced by the nalu event collector library.
Valid Values	`debug`, `info`, `warn`, `error`, `critical`, or `off`
Suggested Value	`info`

Board Controller Logging Level

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/logging/board_controller_log_level`
Description	Log level for messages produced by the nalu board controller library.
Valid Values	`debug`, `info`, `warn`, `error`, `critical`, or `off`
Suggested Value	`info`

Board Python Logging Level

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/logging/board_python_log_level`
Description	Log level for messages produced by the embedded Python board-control layer.
Valid Values	`debug`, `info`, `warn`, `error`, `critical`, or `off`
Suggested Value	`info`

MIDAS Spdlog Max Rate

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/logging/midas_spdlog_max_rate_hz`
Description	Maximum rate in Hz for forwarding spdlog messages into MIDAS messages.
Valid Values	Any positive integer.
Suggested Value	`10`

6.2.4 Midas Parameters

Midas Data Bank Prefix

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/midas_params/data_bank_prefix`
Description	The 2 letter prefix for MIDAS data banks containing event data.
Valid Values	Any 2 letter string.
Suggested Value	`AD`

Note: MIDAS bank names are 4 characters. The first 2 characters are this prefix and the second 2 are used for indexing. So if there is 1 frontend, the data may be stored in `AD00`.

Webpage Initializing Frontend Status Color

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/midas_params/init_color`
Description	Color the frontend status light will use while the frontend is initializing.
Valid Values	Any valid MIDAS color string or color value accepted by the UI.
Suggested Value	`#8A2BE2`

Minimum Bytes to Trigger On

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/midas_params/min_bytes_to_trigger_on`
Description	Minimum number of bytes that must be present in the UDP buffer before the MIDAS thread will trigger collection.
Valid Values	Any non-negative integer.
Suggested Value	`1000`

Note: Higher values may improve throughput at the cost of events being collected in larger batches.

Polling Interval in Microseconds

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/midas_params/polling_interval_us`
Description	Minimum number of microseconds between two consecutive polling cycles.
Valid Values	Any non-negative integer.
Suggested Value	`1000`

Webpage Ready Status Color

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/midas_params/ready_color`
Description	Color the frontend status light will use while the frontend is ready.
Valid Values	Any valid MIDAS color string or color value accepted by the UI.
Suggested Value	`greenLight`

Midas Timing Bank Prefix

Field	Description
Path	`/Equipment/HDSoC-{frontend #}/Settings/midas_params/timing_bank_prefix`
Description	The 2 letter prefix for MIDAS data banks containing timing information.
Valid Values	Any 2 letter string.
Suggested Value	`AT`

Note: MIDAS bank names are 4 characters. The first 2 characters are this prefix and the second 2 are used for indexing. So if there is 1 frontend, timing information may be stored in `AT00`.

Last update: May 22, 2026

7. Scripts

7.1 Top Level

7.1.1 build.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/build.sh`
Description	Builds the project using CMake. It loads the repository environment first, then configures and builds the project.
Flags	`-o, --overwrite` → Remove the existing build directory before building. `--venv PATH` → Override the repo-local virtual environment path. `--system-python` → Ignore the repo-local virtual environment and use the system Python setup.
Example Usage	`./scripts/build.sh` `./scripts/build.sh --overwrite`

7.1.2 run.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/run.sh`
Description	Runs the `frontend` executable. It can build first, set up the repo-local virtual environment, and run under `gdb`.
Flags	`--build` → Build before running. `--overwrite` → Rebuild from a clean build directory when used with `--build`. `--setup-venv` → Create or update the repo-local Python environment before running. `--venv PATH` → Override the repo-local virtual environment path. `--system-python` → Skip the repo-local virtual environment and use the system Python setup. `--debug` → Run the executable in GDB. `-i <number>` → Specifies an index (default: `0`).

Example Usage	`./scripts/run.sh -i 0`
	`./scripts/run.sh --build -i 0`
	`./scripts/run.sh --setup-venv -i 0`

7.2 Data management

7.2.1 delete_data.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/data_management/delete_data.sh`
Description	Deletes MIDAS data files in the experiment directory. If `--dry-run` is used, it lists files without deleting them.
Flags	`--dry-run` → Lists files that would be deleted without removing them.
Example Usage	`./scripts/data_management/delete_data.sh` `./scripts/data_management/delete_data.sh --dry-run`

7.2.2 find_data_dir.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/data_management/find_data_dir.sh`
Description	Finds the experiment data directory from `MIDAS_EXPTAB` and `MIDAS_EXPT_NAME`, then writes it to `experiment_dir.txt`.
Flags	*(None)*
Example Usage	`./scripts/data_management/find_data_dir.sh`

7.3 Environment Setup

7.3.1 Overview

The old `scripts/environment_setup` helpers have been replaced by a newer environment system under:

```
scripts/environment/
scripts/environment/helpers/
```

The main user entrypoint is:

```
source ./scripts/environment/activate_environment.sh
```

This helper will detect and load MIDAS environment variables, optionally create the repo-local Python virtual environment, and activate it.

7.3.2 activate_environment.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/environment/activate_environment.sh`
Description	Main shell entrypoint for preparing the ATAR DAQ environment. It detects and saves MIDAS-related environment variables if needed, creates or refreshes the repo-local virtual environment if needed, and activates it.
Flags	`--quiet` → Suppress informational output. `--no-midas-bin` → Do not prepend `MIDASSYS/bin` to `PATH`. `--system-python` → Skip the repo-local virtual environment and use the system Python setup. `--venv PATH` → Override the virtual environment path.
Example Usage	`source ./scripts/environment/activate_environment.sh`

7.3.3 detect_midas_environment.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/environment/helpers/detect_midas_environment.sh`
Description	Searches for the `midas` directory and `exptab` file, sets `MIDASSYS`, `MIDAS_EXPTAB`, and `MIDAS_EXPT_NAME`, and saves them to `scripts/environment/helpers/environment_variables.sh`.
Flags	`--output PATH` → Override the output file path. `-q, --quiet` → Suppress the summary output.
Example Usage	`./scripts/environment/helpers/detect_midas_environment.sh`

7.3.4 load_environment.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/environment/helpers/load_environment.sh`
Description	Loads the saved repository environment variables and optionally configures access to the repo-local virtual environment. This helper is used internally by `build.sh`, `run.sh`, and other scripts.
Flags	`--quiet` → Suppress informational output. `--add-midas-bin` → Prepend `MIDASSYS/bin` to `PATH`. `--no-venv` → Do not activate or expose the repo-local virtual environment. `--setup-venv` → Create or update the repo-local virtual environment before loading it. `--system-python` → Ignore the repo-local virtual environment and isolate from user-site Python. `--venv PATH` → Override the virtual environment path.
Example Usage	`source ./scripts/environment/helpers/load_environment.sh --add-midas-bin`

7.3.5 setup_venv.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/environment/helpers/setup_venv.sh`
Description	Creates or updates the repo-local Python environment used by board-controller-backed tools. It installs the requirements from `requirements/board_tools.txt`.
Flags	`--venv PATH` → Override the virtual environment path. `--python BIN` → Python executable to use for venv creation. `--requirements PATH` → Requirements file to install. `--recreate` → Delete and recreate the virtual environment.
Example Usage	`./scripts/environment/helpers/setup_venv.sh` `./scripts/environment/helpers/setup_venv.sh --recreate`

7.3.6 activate_venv.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/environment/helpers/activate_venv.sh`
Description	Activates only the repo-local virtual environment.
Flags	*(None)*
Example Usage	`source ./scripts/environment/helpers/activate_venv.sh`

7.3.7 clear_environment.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/environment/helpers/clear_environment.sh`
Description	Clears the MIDAS, ATAR DAQ, and Python environment variables set by the environment helpers.
Flags	*(None)*
Example Usage	`source ./scripts/environment/helpers/clear_environment.sh`

7.4 ODB

7.4.1 set_enabled_channels.py

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/odb/set_enabled_channels.py`
Description	Updates the enabled channels in the ODB by setting the number of enabled channels from the start (`0`) and disabling the rest.
Flags	`num_enabled_channels` → The number of enabled channels, between `0` and `32`.
Example Usage	`./scripts/odb/set_enabled_channels.py 16` Enables the first 16 channels and disables the rest.

7.5 Screen Control

7.5.1 screen_frontend.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/screen_control/screen_frontend.sh`
Description	Starts the frontend inside a detached `screen` session.
Flags	`-i` → Index for the session name (defaults to `0`). `--setup-venv` → Create or update the repo-local virtual environment before running. `--venv PATH` → Override the virtual environment path. `--system-python` → Skip the repo-local virtual environment and use the system Python setup.
Example Usage	`./scripts/screen_control/screen_frontend.sh -i 1`

7.5.2 stop_screen.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/screen_control/stop_screen.sh`
Description	Stops a running `screen` session specified by the index.
Flags	`-i` → Index for the session name (defaults to `0`).
Example Usage	`./scripts/screen_control/stop_screen.sh -i 1`

7.6 Webpage Scripts

7.6.1 start_midas_webpage.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/webpage_scripts/start_midas_webpage.sh`
Description	Starts processes in the background, each inside a `screen` session, using process names from a `screen_names.txt` file.
Flags	None.
Example Usage	`./scripts/webpage_scripts/start_midas_webpage.sh`

7.6.2 stop_midas_webpage.sh

Field	Description
Path	`$ATAR_DAQ_DIR/scripts/webpage_scripts/stop_midas_webpage.sh`
Description	Stops processes running in `screen` sessions based on names listed in `screen_names.txt`.
Flags	None.
Example Usage	`./scripts/webpage_scripts/stop_midas_webpage.sh`

Last update: May 22, 2026

8. Debugging Common Errors

Last update: March 25, 2025

9. Miscellaneous

9.1 Additional Notes

If you're feeling desperate (or perhaps lucky), you can sift through Jack Carlton's work notes. **I warn you that these are not well organized and contain lots of information not about this DAQ**. However, they do contain some documentation of my assembly and troubleshooting of this DAQ.

9.2 Initialism Cheatsheet

See the g-2 modified DAQ's initialism cheatsheet, many of the initialisms are still applicable to this DAQ.

9.3 Networking Tutorial

See the g-2 modified DAQ's networking tutorial for basics. There are many additional (and probably better) online resources as well.

9.4 Useful Midas Information

See the g-2 modified DAQ's midas information page for some general midas tips. There is much more information available on TRIUMF's Midas Wiki page.

9.5 Getting Access to the PIONEER repository

See the g-2 modified DAQ manual's accessing the pioneer repository section and setting up a github ssh token section.

9.6 Port Forwarding an SSH Connection

See the g-2 modified DAQ's port forwarding an SSH connection section.

9.7 Using Screens in Linux

See the g-2 modified DAQ's using screens in linux page.

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